Content Distribution Network (CDN)

1. Overview

Initially, static files (JS, CSS, images, videos) are served directly from your backend (Node.js).

Problem:

- Increased backend load
- High latency for distant users
- Poor scalability

Solution:

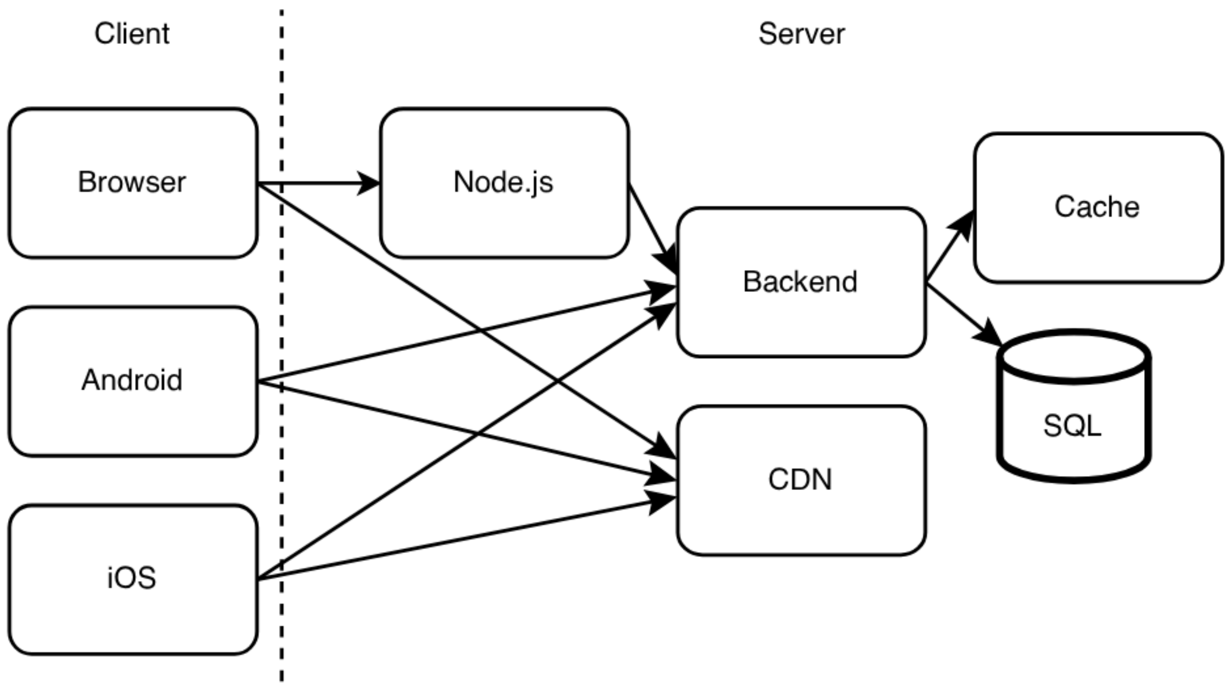
> Use a **Content Distribution Network (CDN)** to serve static content from geographically distributed servers

2. What is a CDN?

A **CDN (Content Distribution Network)** is a network of globally distributed servers that:

> **cache and deliver static content from locations closest to users**

3. Updated Architecture



4. Request Flow

Static Content (via CDN):

1. Client requests static file (e.g., JS, image)
2. Request goes to CDN
3. CDN serves from **nearest edge server**

Dynamic Content:

1. Client → Backend
2. Backend → Cache / DB

5. What Goes to CDN?

Static Content:

- JavaScript files
- CSS
- Images
- Videos
- Fonts

Not Suitable:

- Dynamic API responses
- Personalized data

6. How CDN Works

Step 1:

Files uploaded to CDN

Step 2:

CDN replicates files across **global edge locations**

Step 3:

User requests file → served from nearest location

7. CDN vs GeoDNS

Feature	CDN	GeoDNS
Purpose	Content delivery	Request routing

Used for	Static files	Backend routing
Level	Edge servers	DNS level

Both work together in large systems

8. Benefits of CDN

1. Reduced Latency

- Content served from nearest location

2. Reduced Backend Load

- Static content offloaded

3. Improved Throughput

- Faster downloads

4. High Availability

- Multiple edge locations

5. Cost Optimization

- Reduced server bandwidth usage

9. Advanced Concepts

Edge Caching

- Content stored at edge servers

Cache TTL

- Controls how long content stays cached

Cache Invalidation

- Remove outdated content

Example:

- New version of JS file → invalidate old cache

Cache Busting

- Change file name/version
  - `app.js → app.v2.js`

10. Challenges

1. Cache Invalidation

- Hard to ensure latest content

2. Dynamic Content Handling

- CDNs not ideal for personalized responses

3. Cost at Scale

- Large CDN usage can be expensive

11. Evolution from Previous System

Before:

- Backend served everything

Now:

- CDN serves static content
- Backend handles only dynamic requests

Huge performance improvement

12. Real-World CDN Providers

- Cloudflare
- Amazon CloudFront
- Rackspace

13. Final Summary

- CDN = distributed system for static content delivery
- Reduces latency and backend load
- Uses global edge servers
- Works best for **static assets**
- Requires **cache invalidation strategy**